

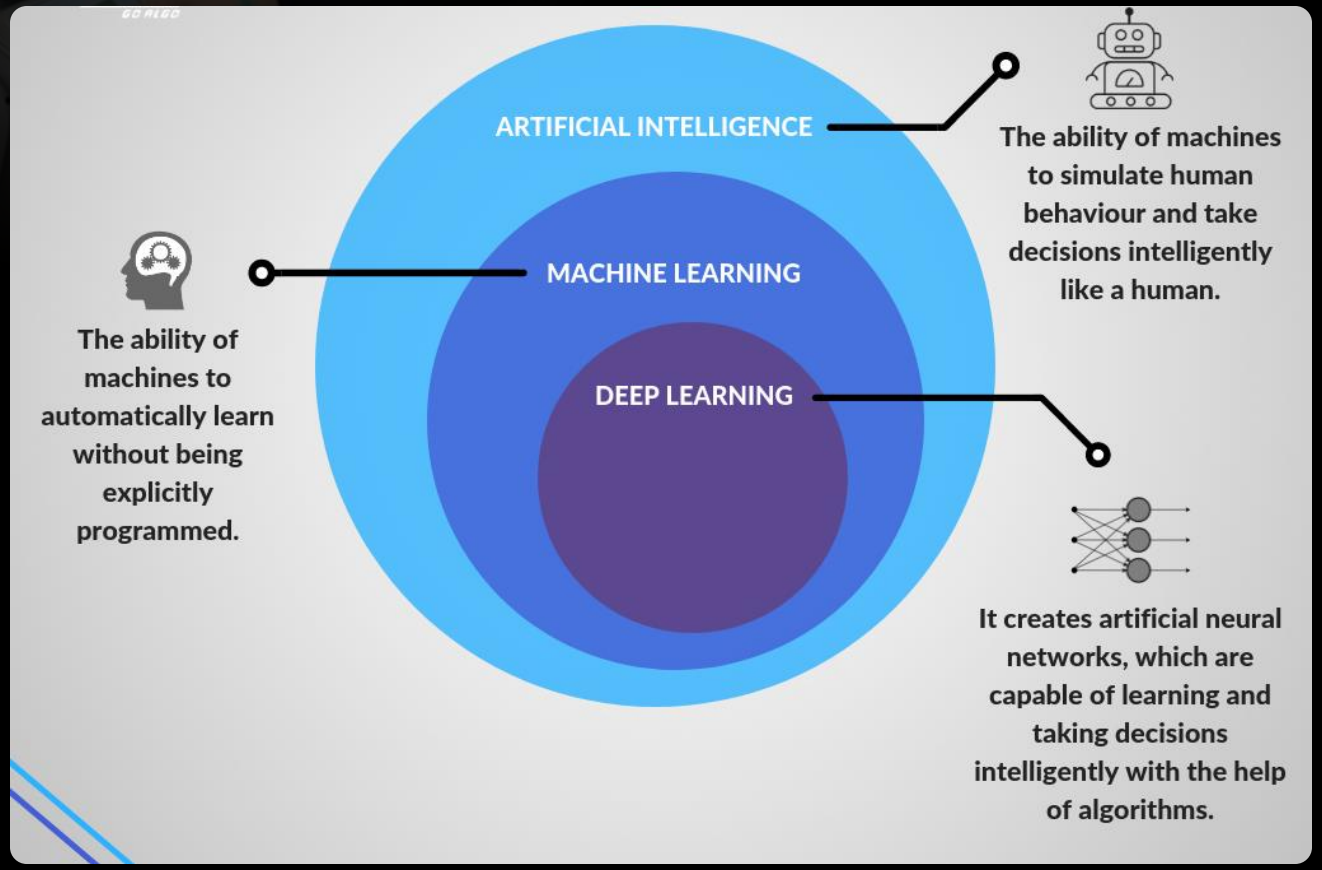
COMPUTER VISION (TFP505)

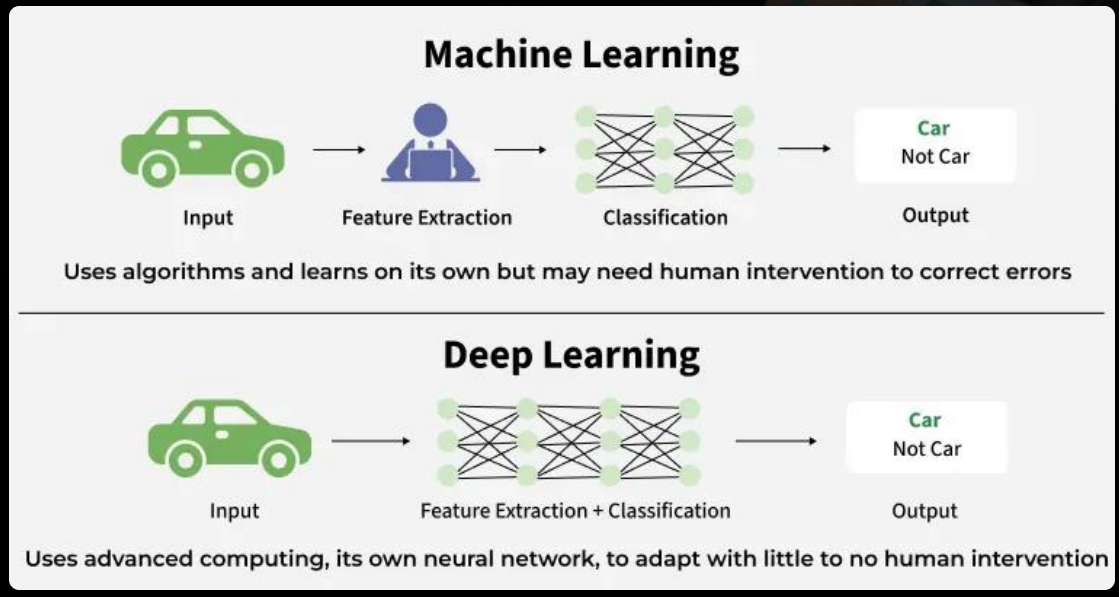
Pertemuan 9 – Basic Deep Learning

ALIFIA REVAN PRANANDA

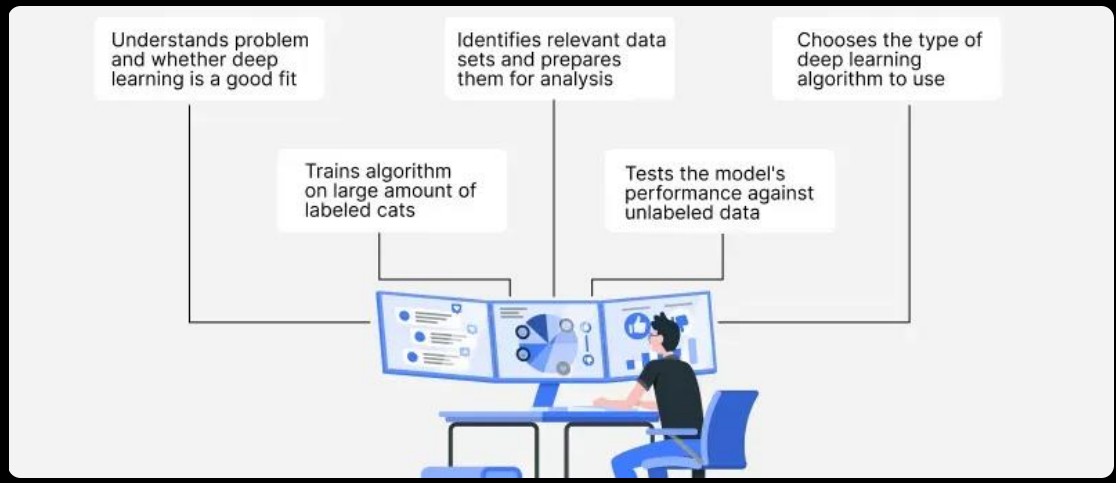
Department of Information Technology
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DEEP LEARNING

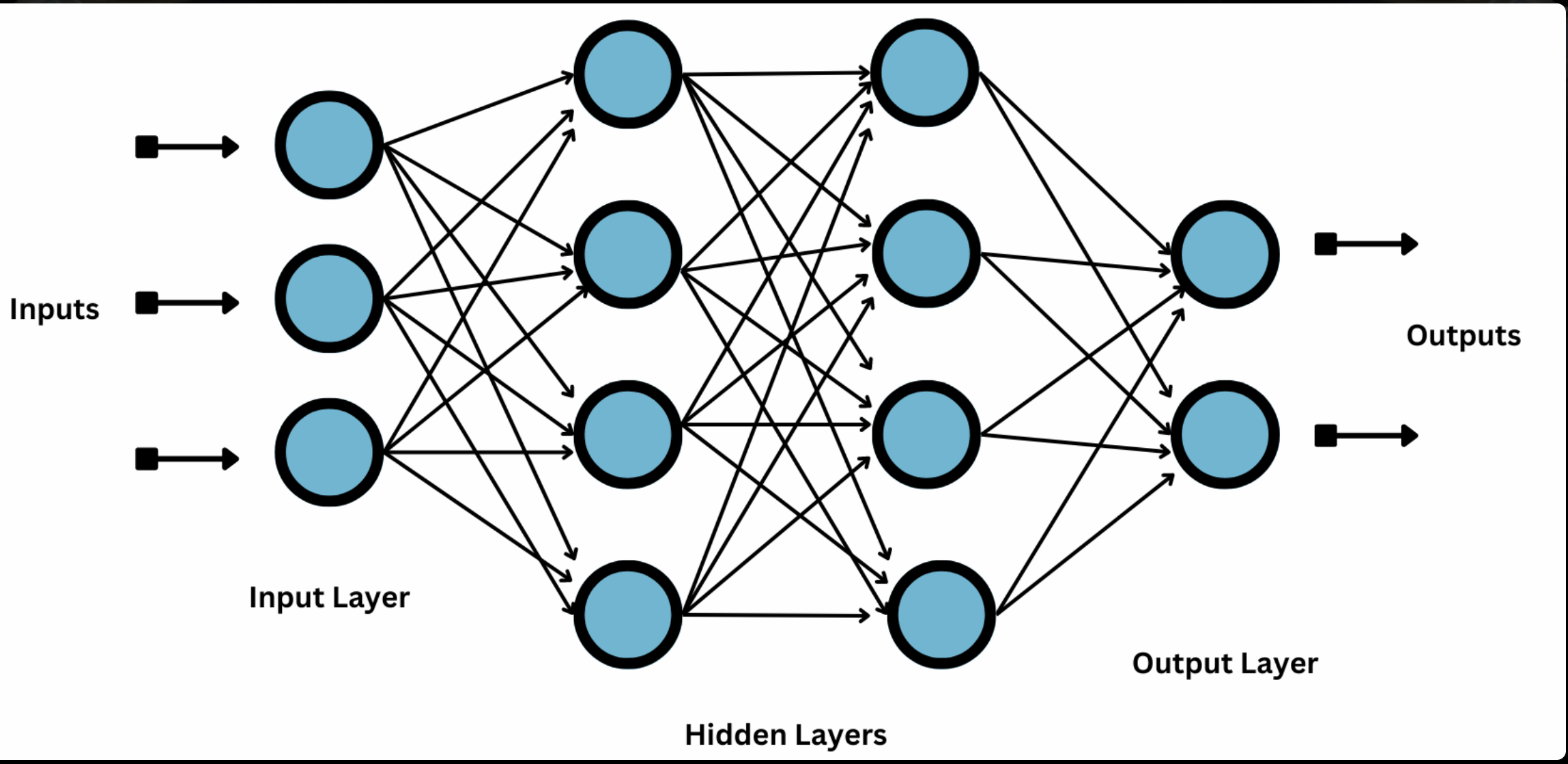




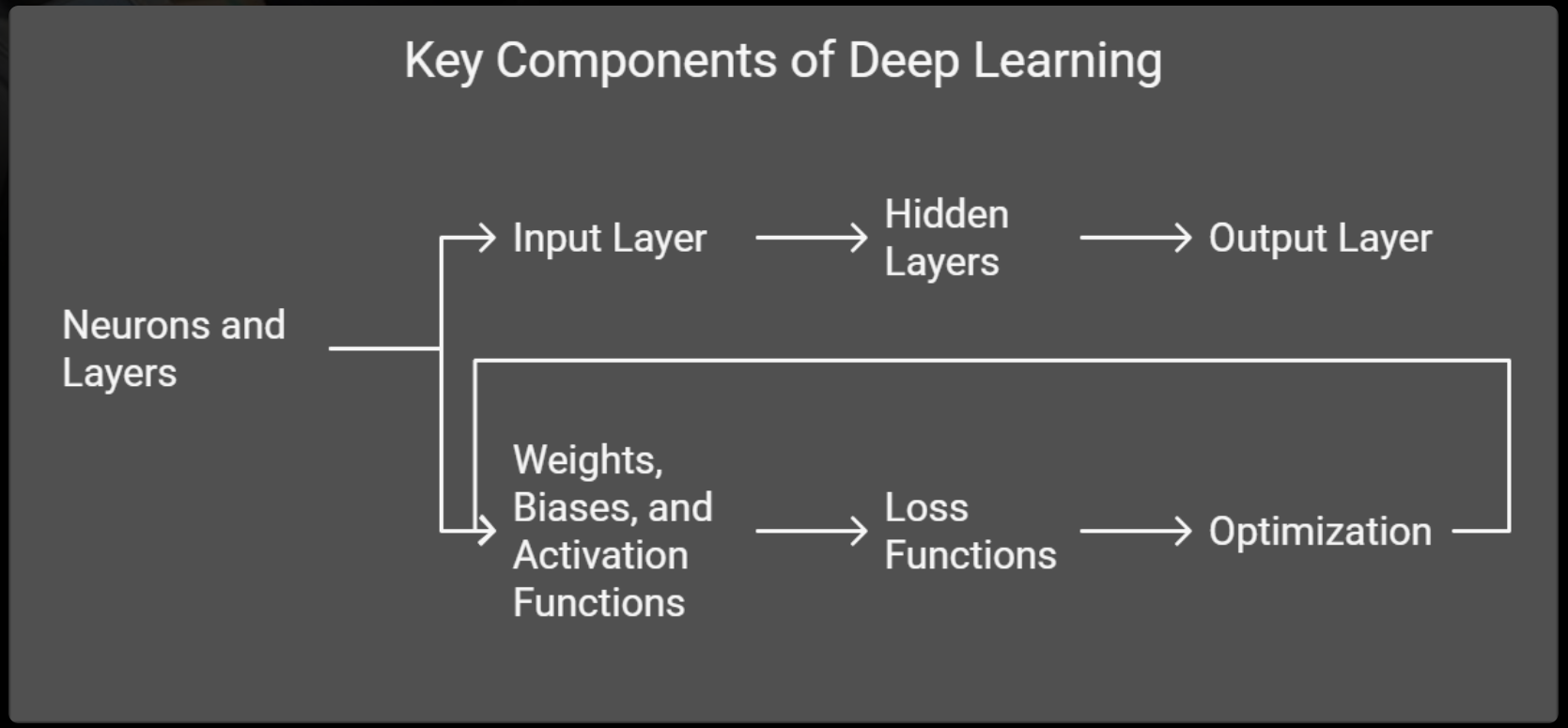
DEEP LEARNING is a subset of **machine learning** that focuses on using **artificial neural networks**—especially those with many layers (hence "deep")—to learn patterns and representations from large amounts of data.



DEEP LEARNING



DEEP LEARNING

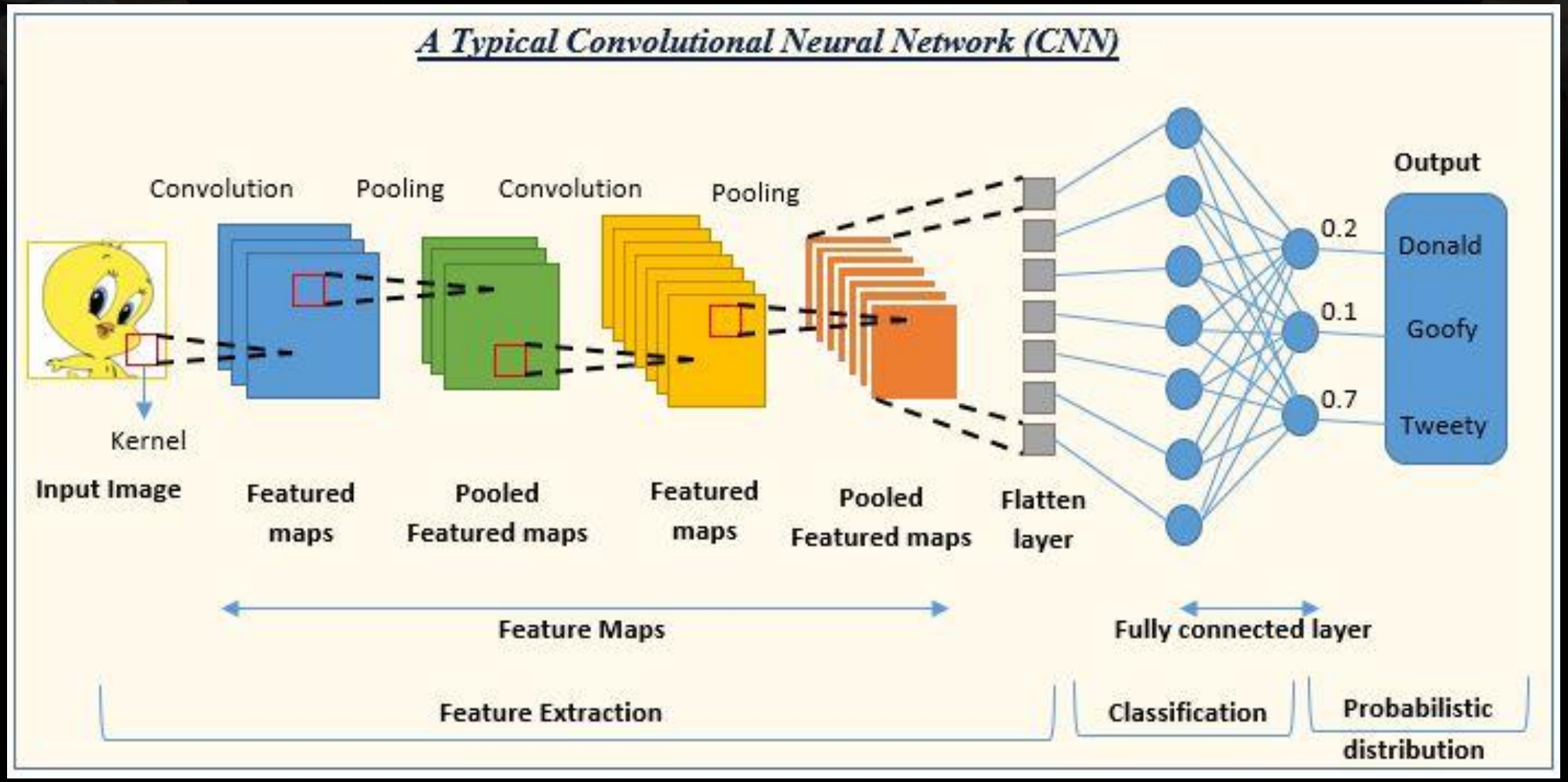


- Learning process: The model adjusts its internal parameters (weights and biases) through backpropagation and gradient descent to minimize error.
- Data requirement: It typically requires large datasets and high computational power (GPUs/TPUs).
- Applications: Used in computer vision, speech recognition, natural language processing, autonomous vehicles, healthcare, and more.

Key Points:

Architecture: Deep learning models are built from multiple layers of neurons—input, hidden, and output layers—that transform data step by step to extract features and make predictions.

CNN


A Typical Convolutional Neural Network (CNN)
Convolution
Pooling
Convolution
Pooling
Output
0.2
0.1
0.7
Donald
Goofy
Tweety
Kernel
Input Image
Featured maps
Pooled Featured maps
Featured maps
Pooled Featured maps
Flatten layer
Feature Maps
Fully connected layer
Feature Extraction
Classification
Probabilistic distribution

TUGAS

1. Cari data image classification di Kaggle
2. Buat arsitektur deep learning sederhana untuk kasus tersebut
3. Lakukan evaluasi dengan menghitung accuracy, precision, recall dan F1-score